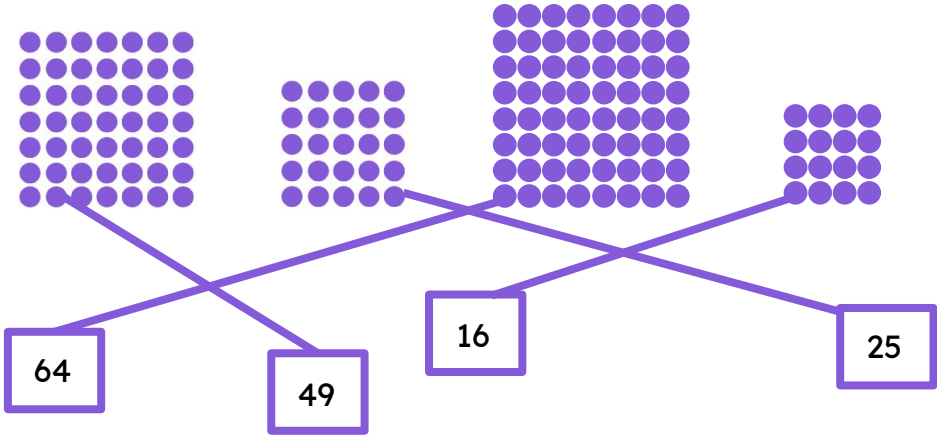




Square and cube numbers

Task A: Square numbers

1) Match up the arrays with the correct square number.



2) Work out the age given the clues.

My teacher's age is:
a square number,
a multiple of 3,
and less than 70.

How old is the teacher?

Teacher's age is 36

- 1
- 4
- 9
- 16
- 25
- 36
- 49
- 64

Alex is a student at Oak Academy.
Alex's age is:
4 less than a square number and is
an even number.

How old is Alex?

Alex's age is 12

3) Identify if the following statements are true or false. Justify your answer.

Statement	True	False	Justification
121 is a square number	✓		$11 \times 11 = 121$
$13 \times 13 = 169$ so 13 is a square number		✓	169 is the square number
4 add 3 then multiply by 7 gives a square number	✓		$4+3 = 7$ then $7 \times 7 = 49$ 49 is a square number
7 multiplied by 4 then add 3 gives a square number		✓	$7 \times 4 = 28$ then $28 + 3 = 31$ 31 is not a square number

4) Jun discovers he can sum two different square numbers to make a third square number.

a) Can you find the missing square numbers to complete this table?

1st square number	2nd square number	Resultant square number
9	16	25
36	64	100
25	144	169

b) Can you find another set of two different square numbers to make a third square number? *There are many out there. E.g. $81+144=225$ and $64+225=289$*

5) How many squares can you see on an 8 by 8 chess board?

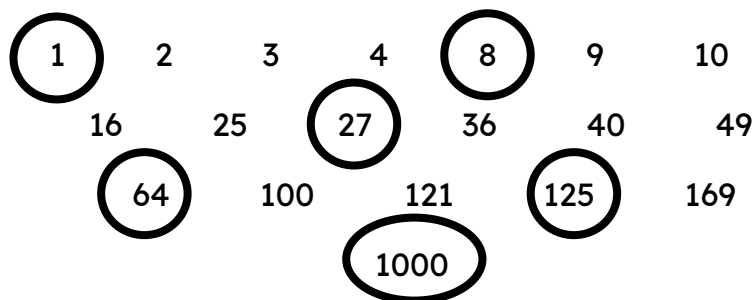
204 squares.

This is because you have to calculate how many 1×1 squares, 2×2 square, 3×3 squares and so on.

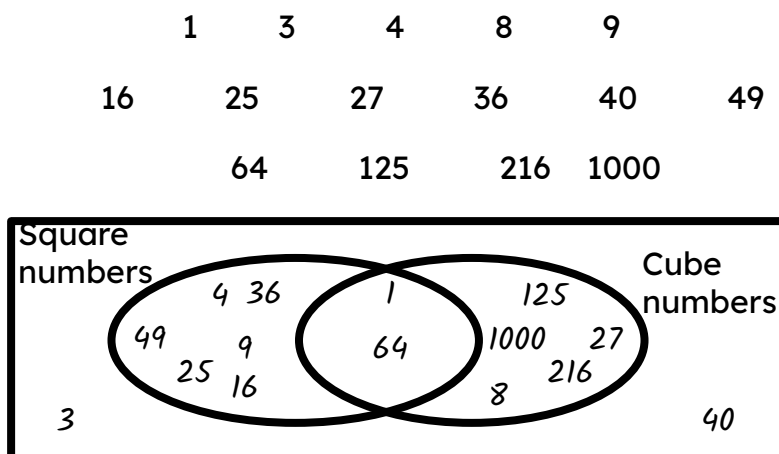
These numbers end up being the square numbers: 64, 49, 36, 25, 16, 9, 4, 1. Summing these gives 204 squares on a chessboard,

Task B: Cube numbers

1) Identify all the cube numbers from the list below.



2) Put the following numbers in the Venn diagram:



3) Fill in the number to make the calculation correct.

The number in the SQUARE is a square number

The number in the CUBE is a cube number.

- a) $\square + \text{cube} = 31$
- b) $\square + \text{cube} = 33$
- c) $\square + \text{cube} = 225$
- d) $\square + \square + \text{cube} = 104$
- e) $\text{cube} + \text{cube} + \text{cube} = 36$

- a) $4 + 27 = 31$
- b) $25 + 8 = 33$
- c) $100 + 125 = 225$
- d) $36 + 4 + 64 = 104$
- e) $1 + 8 + 27 = 36$

